GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

# Pythagorean Theorem

*Independent Set 6.03* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A right triangle has legs of 12 m and 5 m. How long is the hypotenuse?

***13 m***

*144 + 25 = 169 = $13^2$.*

**2.** A right triangle has legs of 3 ft and 8 ft. How long is the hypotenuse, exactly and to the nearest tenth?

***√73 ft ≈ 8.5 ft***

*9 + 64 = 73.*

**3.** A right triangle has a hypotenuse of 39 cm and one leg of 36 cm. How long is the other leg?

***15 cm***

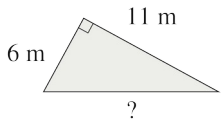
*1,521 − 1,296 = 225 = $15^2$.*

**4.** A right triangle has a hypotenuse of 11 inches and one leg of 4 inches. How long is the other leg, exactly and to the nearest tenth?

***√105 in ≈ 10.2 in***

*121 − 16 = 105.*

**5.** The right triangle on your paper has legs of 6 m and 11 m. How long is its hypotenuse, to the nearest tenth?




***√157 m ≈ 12.5 m***

*36 + 121 = 157; c is across from the square corner.*

**6.** A right triangle has a hypotenuse of 25 yd and one leg of 24 yd. The other leg is x. Write the equation that finds x, then find x.

***$24^2 + x^2 = 25^2$; x = 7 yd***

*625 − 576 = 49 = $7^2$.*